

Active Learning-DETR: Cost-Effective Object Detection for Kitchen Waste

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Abstract—Object detection in kitchen waste faces numerous challenges, including a variety of target categories, significant morphological variations, and complex backgrounds, coupled with the scarcity of targeted datasets, resulting in limited research in this area. In response to these challenges, we integrate insights from real-world industrial scenarios of kitchen waste sorting and create, for the first time, a dataset comprising eight categories specifically designed for kitchen waste object detection. Furthermore, we initiate the integration of active learning (AL) methods into the object detection transformer (DETR) network, proposing the AL-DETR model—a high-precision detection model tailored for scenarios with limited samples. To overcome the drawback of DETR, which relies on a large amount of annotated data for training, we employ an AL strategy, annotating challenging samples with high uncertainty selected from a pool of unlabeled data, and integrating them back into the training set to enhance the model’s learning capability. AL-DETR also models non-local visual correlations between labeled and unlabeled samples, comparing differences in detection categories to identify relevant unlabeled samples. Additionally, we establish a robotic sorting platform with an end-effector integrating gripping and suction functionalities, accompanied by the development of visual detection and robot control software. The experimental results indicate that, in the dataset constructed for this study, which primarily focuses on detecting large objects, our approach achieves a minimum improvement of 9.23% compared to other methods, reaching state-of-the-art (SOTA) performance. Additionally, our method surpasses other AL approaches using the entire annotated dataset by more than 1.01% in mAP when the original dataset’s annotation data is only 82.5%. In practical testing, the sorting platform successfully accomplishes efficient detection and sorting of kitchen waste.

Index Terms—Active learning (AL), kitchen waste images, object detection, robotic sorting, transformer.

Manuscript received 24 October 2023; revised 11 January 2024; accepted 24 January 2024. Date of publication 22 February 2024; date of current version 28 February 2024. This work was supported in part by the Project of the National Key Research and Development Program of China under Grant 2021YFC1910402, in part by the National Natural Science Foundation of China under Grant 62073129 and Grant U21A20490, in part by the Hunan Provincial Natural Science Foundation of China under Grant 2022JJ10020, and in part by the Scientific Research Project of Hunan Education Department under Grant 21B0330. The Associate Editor coordinating the review process was Dr. Jae-Ho Han. (*Corresponding author: Qiaokang Liang.*)

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Digital Object Identifier 10.1109/TIM.2024.3368494

I. INTRODUCTION

WASTE management is a global challenge, with billions of tons of waste generated daily worldwide, incurring substantial disposal costs. Compared to traditional landfill and incineration methods, waste sorting, recycling, and reprocessing align more closely with the recent advocacy for green, low-carbon societal development. In China, waste is broadly categorized into four types: kitchen waste, recyclables, hazardous waste, and other waste [1]. Kitchen waste, comprising vegetables, fruits, leftovers, peels, and bones generated from daily life, food manufacturing, and catering services, is the primary focus of urban waste management. According to the Food and Agriculture Organization of the United Nations, global kitchen waste production exceeds 1.3 billion tons annually [2]. In 2020, China’s kitchen waste production was estimated to be around 72.44–127.4 million tons [3], as per the “China Statistical Yearbook (2020).” Due to factors such as lax waste disposal practices, kitchen waste often includes plastics, glass, fibers, metals, and biomass, posing new challenges for waste sorting. Fortunately, amid the rapid development and widespread adoption of artificial intelligence, leveraging deep learning technologies to assist in waste sorting [4], [5], [6], [7] not only replaces manual sorting in adverse conditions but also significantly enhances the accuracy and precision of waste detection and classification. Therefore, constructing a precise system for the identification, detection, and classification of kitchen waste is of paramount research significance and practical value. Such a system facilitates the recycling of valuable waste, including plastic bottles, aluminum cans, and tetra paks, while simultaneously removing non-biodegradable items such as rigid paper, and other plastic production from kitchen waste. The residual waste consists of biodegradable organic matter. This endeavor holds immense importance in advancing the development of a green circular economy.

In recent years, deep learning has seen numerous successful applications in image-based waste object detection. Contemporary deep learning technologies have strengthened high-performance visual recognition models, advancing the automation of waste classification tasks [8]. Fang et al. [9] introduced an instance-balanced detector tailored for intelligent detection and classification of kitchen waste. Li et al. [10] introduced a multimodel cascaded convolutional neural network (MCCNN) for the detection and classification of municipal solid waste images, addressing the inefficiency associated with relying on manual efforts for the detection and classification of household waste. Jin et al. [11] enhanced the

MobileNetV2 deep learning model for garbage detection and classification, while Majchrowska et al. [12] improved deep network models by employing EfficientDet-D2 for garbage localization and EfficientNet-B2 for detection and classification, achieving state-of-the-art (SOTA) performance in garbage object detection. Recently, transformers have demonstrated superior performance in the field of object detection compared to convolutional neural networks. The introduction of transformers into object detection, notably the end-to-end detection transformer (DETR) model [13] proposed by Carion et al., has surpassed other CNN methods in both detection accuracy and runtime performance. Subsequent improvements to DETR, such as Conditional DETR [14], Dynamic DETR [15], and Deformable DETR [16], have achieved SOTA results on public datasets like COCO. Transformer models have also found success in practical scenarios, as seen in the application of the PA-DETR [17] method for detecting visually indistinguishable bolt defects in power transmission lines. However, there has been limited research on employing transformers for kitchen waste detection. In light of this, we have enhanced the DETR model and, for the first time, applied it to our constructed dataset for kitchen waste object detection, spanning eight distinct categories.

In the realm of computer vision, it is well-established that deep learning models demand a substantial volume of annotated data for effective training. However, the practical challenges of acquiring labeled data and the frequent impracticality of annotating all available data are recognized. In recent years, many researchers have conducted studies on training deep neural network models using a combination of partially annotated data and a large volume of unlabeled data, with active learning (AL) being one of the methodologies employed. AL-based approaches have shown considerable success in various applications, including traditional machine learning [18], [19], [20] as well as deep learning tasks such as image classification [21], segmentation [22], and detection [23]. In the domain of deep learning-based object detection tasks, Kao et al. [24] proposed metrics to assess the informativeness of assumed objects, employing AL to curtail the amount of annotated data required to achieve target object detection performance. Wang et al. [25] introduced a principled self-supervised sample mining (SSM) process, overcoming the limitation of AL's current definition of sample selection criteria within a single-image environment. This addresses suboptimal robustness and unsuitability for large-scale object detection problems. In a paper by [26], variational autoencoders (VAEs) and adversarial networks were utilized to learn a latent space capable of distinguishing between labeled and unlabeled data. However, our research indicates a current absence of methods integrating AL with DETR, particularly in practical applications for kitchen waste detection and classification. Therefore, this article proposes an AL-based transformer object detection method. In this approach, a substantial amount of unlabeled data captured in waste processing facilities is leveraged, and AL is employed to select data with the highest uncertainty for annotation. This method ensures high detection accuracy while simultaneously reducing the cost of annotated data, offering significant practical value.

A comprehensive system, in addition to robust algorithms, relies on the support of both data and terminals. While there is an abundance of datasets available for general waste [27], [28], the specific niche of datasets dedicated to kitchen waste object detection remains limited. Notably, Fang et al. [9] introduced a dataset encompassing four distinct categories—metal, glass, plastic bottles, and tetra paks. However, the kitchen waste received by processing plants often encompasses a diverse array of contaminants, including waste bags, rigid paper, and other plastic production. These impurities wield a significant impact on the subsequent fermentation and processing of kitchen waste. Acknowledging the constraints of existing datasets, particularly their limited number of categories, we embarked on pioneering the creation of an eight-category kitchen waste dataset. The objective of this dataset is to facilitate more precise classification for garbage recycling and processing. At the terminal of the sorting equipment, much like other inspection tasks such as the snake-arm robot system employed for low-light industrial inspection [29] and the surface defect detection system for aero-engine blades based on a flexible robotic arm [30], we established a parallel robot sorting platform. This advanced platform incorporates a mechanical hand with versatile gripping and suction capabilities. Additionally, we engineered XRobot—a comprehensive software suite encompassing visual detection and robot control functionalities.

We focus on the practical application of kitchen waste object detection and classification, covering the entire process from dataset construction to the development of a cost-effective object detection algorithm. It extends to the control software for the sorting platform and ultimately the versatile mechanical hand of the parallel robot for efficient recycling and impurity removal in kitchen waste. The main contributions of this article are summarized as follows.

- 1) Initially, we have constructed an eight-category kitchen waste object detection dataset using authentic images captured on the production line of a kitchen waste processing plant over the course of a year.
- 2) For the first time, we have integrated AL into a Transformer-based object detection network, proposing the AL-DETR model. Leveraging the consistency of predictions between original and augmented images, we utilize this as the measure of uncertainty for AL. The AL-DETR model queries a small set of challenging samples and incorporates them into the DETR module. Detection results are output through the designed query and prediction branch structure. Experimental results demonstrate achieving SOTA kitchen waste object detection performance with only 80% annotated data.
- 3) We have successfully implemented a comprehensive kitchen waste sorting platform, featuring a high-speed parallel robot as its core. The platform incorporates a versatile mechanical hand at the end, capable of both suction and gripping. Additionally, we have devised control software for the sorting platform, facilitating efficient recycling and impurity removal in kitchen waste. The on-site demonstration video of the

robot sorting system is available at: <https://github.com/seedahi/Robotic-Intelligent-Waste-Sorting>.

II. DATASET AND ROBOTIC PLATFORM OVERVIEW

In this section, we initially present the recently developed object detection dataset specifically tailored for kitchen waste, referred to as OD-KW. Subsequently, we present an overview of the module composition and operational principles of the robot sorting platform constructed in this study.

A. Dataset Description

A high-quality kitchen waste detection dataset holds significant potential for industries focused on waste utilization and circular economy development. It aligns seamlessly with the current emphasis on green and low-carbon development. Following our investigation, we identified a gap in the availability of a real industrial application dataset specifically tailored for kitchen waste in practical scenarios. The images in the OD-KW dataset constructed in this study originate from the production line of Huanchuang Technology (Zhengzhou) kitchen waste processing plant. To ensure temporal representativeness and category richness, we gathered 15 994 images over the span of a year, considering variations across different months. It is essential to note that this dataset remains open-ended, allowing for the emergence of new categories beyond the original dataset scope over time.

The original images within the OD-KW dataset maintain a fixed high resolution (i.e., 3256×2724 pixels). To align with the specifications of our hardware platform and facilitate model processing, we uniformly adjusted the image resolution to 640×480 . Considering the diverse waste categories present in kitchen waste, we identified eight distinct targets for bounding box annotation. These objects include waste bags, cans, shoes, plastic bottles, other plastic products, glass bottles, cardboard boxes, and tetra paks, with the exception of biodegradable biomass waste. Fig. 1 presents partial examples of original images from the OD-KW dataset. Fig. 2 illustrates the proportion of targets in each category, and Fig. 3 provides examples and visual representations of the eight different object categories. Each object instance was annotated using a bounding box denoted by the coordinates (x_1, y_1, x_2, y_2) , representing the left-top vertex (x_1, y_1) and the right-bottom point (x_2, y_2) . The dataset was partitioned into training and test sets in an 8:2 ratio.

In our dataset, plastic packaging bags are considered a single category (referred to as “waste bag”). Similarly, aluminum cans, tin cans, metal sheets, and the like are grouped as a single category (referred to as “metal”). Polyethylene terephthalate (PET), polyethylene (PE), polypropylene (PP), and other materials, as well as visually intact plastic bottles, are considered the same category (referred to as “plastic bottle”). Plastic products other than visually intact plastic bottles, such as disposable plastic cups, plastic packaging boxes, and plastic bowls, are considered as a single category (referred to as “other plastic”). Packaging made of rigid paper is classified as a distinct category (referred to as “carton”). Milk cartons, juice boxes, and similar containers are grouped



Fig. 1. Partial examples of original images from the OD-KW dataset.

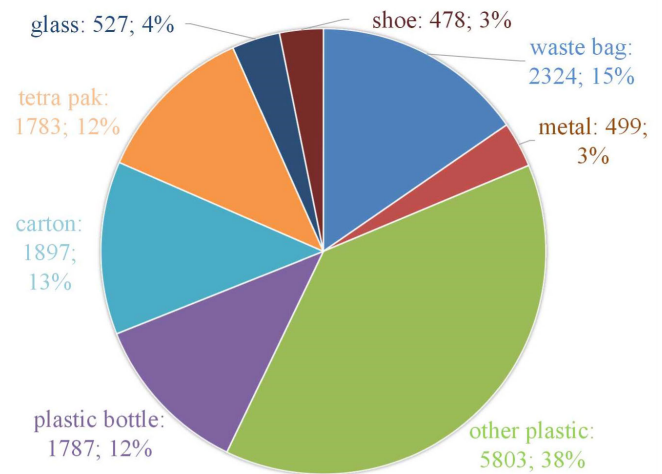


Fig. 2. Quantities and proportions of different object categories in the OD-KW dataset.

together as one category (referred to as “tetra pak”). Relatively intact glass bottles form a separate category (referred to as “glass”). Additionally, it is noteworthy that in our kitchen waste dataset, we observed a notable presence of various types of shoes mixed with the waste. Hence, we have designated shoes as a distinct category (referred to as “shoe”).

B. Robotic Detection and Sorting Platform

We designed and constructed the kitchen waste sorting robot to validate the model’s generalization capability proposed in this article and to establish a complete processing system for kitchen waste, spanning from feeding to sorting. Illustrated in Fig. 4(a), the kitchen waste sorting robot primarily comprises a vision detection module, a parallel robot, a conveyor belt, and a host computer. Depicted in Fig. 4(b), upstream of the conveyor belt, a camera with a resolution of 1600×1200 and a frame rate of 60 frames/s, accompanied by a lighting source, is mounted upstream of the conveyor belt. The parallel robot, illustrated in Fig. 4(c), is equipped with integrated suction cups

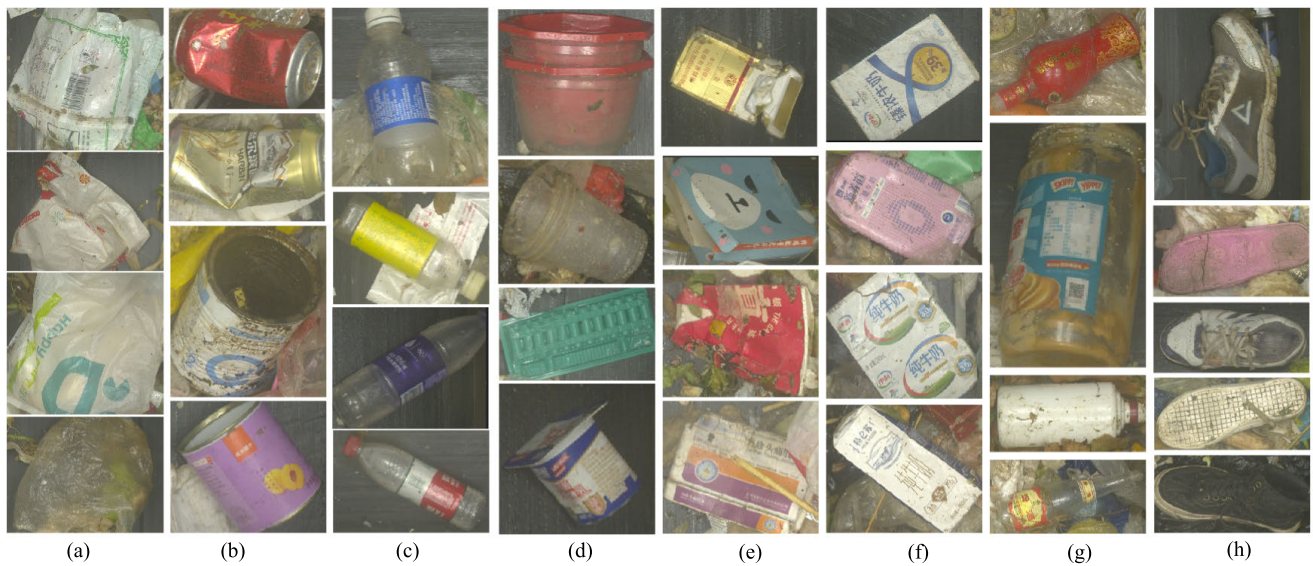


Fig. 3. Annotated objects examples across eight classes [(a)waste bag, (b) metal, (c) plastic bottle, (d) other plastic, (e) carton, (f) tetra pak, (g) glass, and (h) shoe] in the constructed OD-KW.

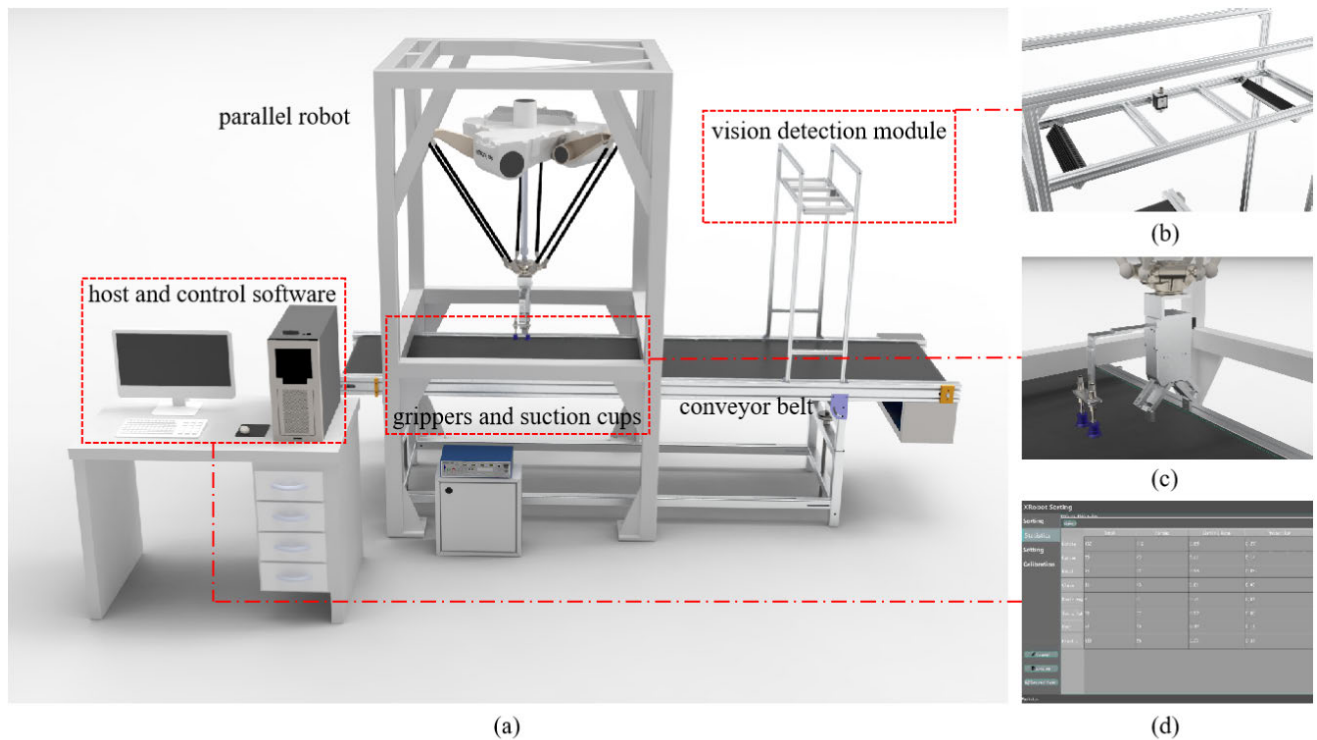


Fig. 4. (a) Schematic representation of the kitchen waste sorting platform. (b) Visual detection module. (c) Functional module for gripping and adhesion at the end-effector of the robotic arm. (d) Control software for detection and sorting, XRobot.

and grippers at the end, employing diverse sorting methods tailored to different targets. Shown in Fig. 4(d), the visual detection and robot control software, XRobot, are integrated on the host computer. AL-DETR serves as the core algorithm of the software, detecting images captured by the vision unit, generating detection frames and predicting categories, and transmitting instructions to the robot for sorting.

The schematic of the robot kitchen waste sorting system constructed in this article is shown in Fig. 5(a). The system consists of the vision unit, processing unit, and execution

unit, responsible for image acquisition, object recognition and control, and implementing object grasping, respectively. The main components of the system are connected to the computer in the processing unit through two buses, Ethernet and RS485. The sorting control flow diagram is illustrated in Fig. 5(b). The specific sorting process is as follows: first, the camera captures kitchen waste images on the conveyor belt at a fixed frequency and sends them to the AL-DETR model for object detection. The detected objects in each image are then filtered by the object filter, which sets object thresholds based on their

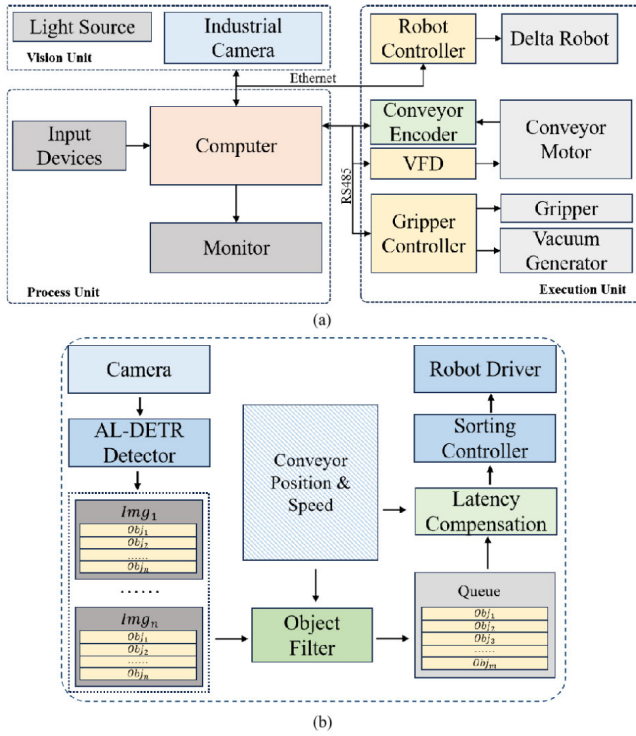


Fig. 5. (a) Schematic of the robot kitchen waste sorting system. (b) Schematic of the sorting control flow.

positions and sizes on the conveyor belt, filtering out duplicate and excessively small objects and outputting them to the object queue. Subsequently, the detected objects undergo position delay compensation based on the conveyor belt speed and system processing delay time before being sent to the sorting controller. This controller sorts the objects, selects appropriate tools (gripper or suction cup) based on the object category, generates grabbing paths sequentially, and finally drives the robot for grasping through the robot driver.

III. METHODOLOGY

This section introduces the AL-DETR model, depicted in Fig. 6. Drawing inspiration from [13], [31], and [32], we opt for DETR as the fundamental model and CALD as our AL strategy. Significantly, this article introduces the integration of AL with the DETR method, a SOTA approach for object detection, incorporating enhancements and innovations.

In particular, we feed the annotated image set, the unlabeled image set along with its augmented images into the CNN (ResNet) backbone to learn the 2-D representation of input images. We extract batch-encoded visual features for each image, flatten and complement them with positional encoding, and then pass them to the transformer encoder. The decoded output is then forwarded to the branching branch, where individual information of unlabeled samples is evaluated through uncertainty-based references and corresponding prediction metrics. Difficult samples are selected for annotation and added to the labeled sample pool. On the other hand, we pass the decoder's output, containing embedded information for each labeled sample, to a shared feedforward network (FFN) with prediction information for object categories and bounding boxes. Algorithm 1 provides the pseudocode for the AL-DETR framework.

A. Feature Learning Network

DETR is a pioneering effort in realizing end-to-end object detection using a transformer. In contrast to a conventional Transformer, DETR streamlines the detection pipeline by eliminating various handcrafted elements that encode prior knowledge, such as spatial anchors or non-maximal suppression. This results in a significantly smaller parameter size, making it conducive to hardware environments and tasks with relatively modest dataset sizes. The architecture comprises four key components: a ResNet backbone for extracting concise feature representations, an encoder, a decoder converter, and an FFN serving as the detection head for the final prediction.

The fundamental concept of AL involves training the learner with a small set of labeled data and strategically selecting samples with the highest uncertainty from a vast pool of unlabeled data for labeling by an oracle (e.g., a human expert). This approach aims to reduce labeling costs. In this study, we maintain an equal proportion of labeled and unlabeled examples to ensure a balanced training dataset. Leveraging the Transformer's feature extraction method, each image is partitioned into patches through a regular grid, treating each patch as an input token for the transformer. We introduce a visual transformer between the feature extractor and the output layer, enhancing the batches of feature maps obtained from the feature extractor. In a batch of B examples, we select labeled images with ground truth (GT) $(x_L, y_L) \in L$, and simultaneously, there are unlabeled images $x_u \in U_B$ in the unlabeled pool U , where $x_u \in U \setminus L$. We input these images to the feature extractor $f: \mathbb{R}^{B \times W \times H \times C} \rightarrow \mathbb{R}^{B \times wh \times |F|}$. Here, W and H represent the height and width of the image, C is the number of channels in the input image (three color channels in this article), w and h denote the width and height of the channels of the last convolution layer, and $|F|$ is the total number of filters in the last convolution layer of the feature extractor. The output of these layers is fed to the encoder before being transmitted to the output layer. While traditional CNN layers primarily handle local dependencies, the Transformer excels in modeling non-local dependencies. In this article, it effectively establishes non-local dependencies between labeled and unlabeled examples, allowing the selection of complementary unlabeled examples for subsequent labeling.

The essential components of the Transformer's encoder include a batch self-attention block and an FFN with residuals and layer normalization. In the self-attention block, we transpose the feature maps to the batch and connect them, represented as $f_T(x; \lambda) \in \mathbb{R}^{B \times wh \times |F|}$ (λ denotes the parameters of the feature extractor). This process constructs output functions that establish interactions between key, query, and value matrices (W_a^K, W_a^Q, W_a^V) to facilitate feature interactions. The operations of batch self-attention are summarized as follows:

$$S_a = f_T(x; \lambda) + \frac{\text{softmax}\left(\frac{(f_T(x; \lambda)W_a^K)(f_T(x; \lambda)W_a^Q)^T}{\sqrt{d_K}}\right)(f_T(x; \lambda)W_a^V)}{\sqrt{d_K}} \quad (1)$$

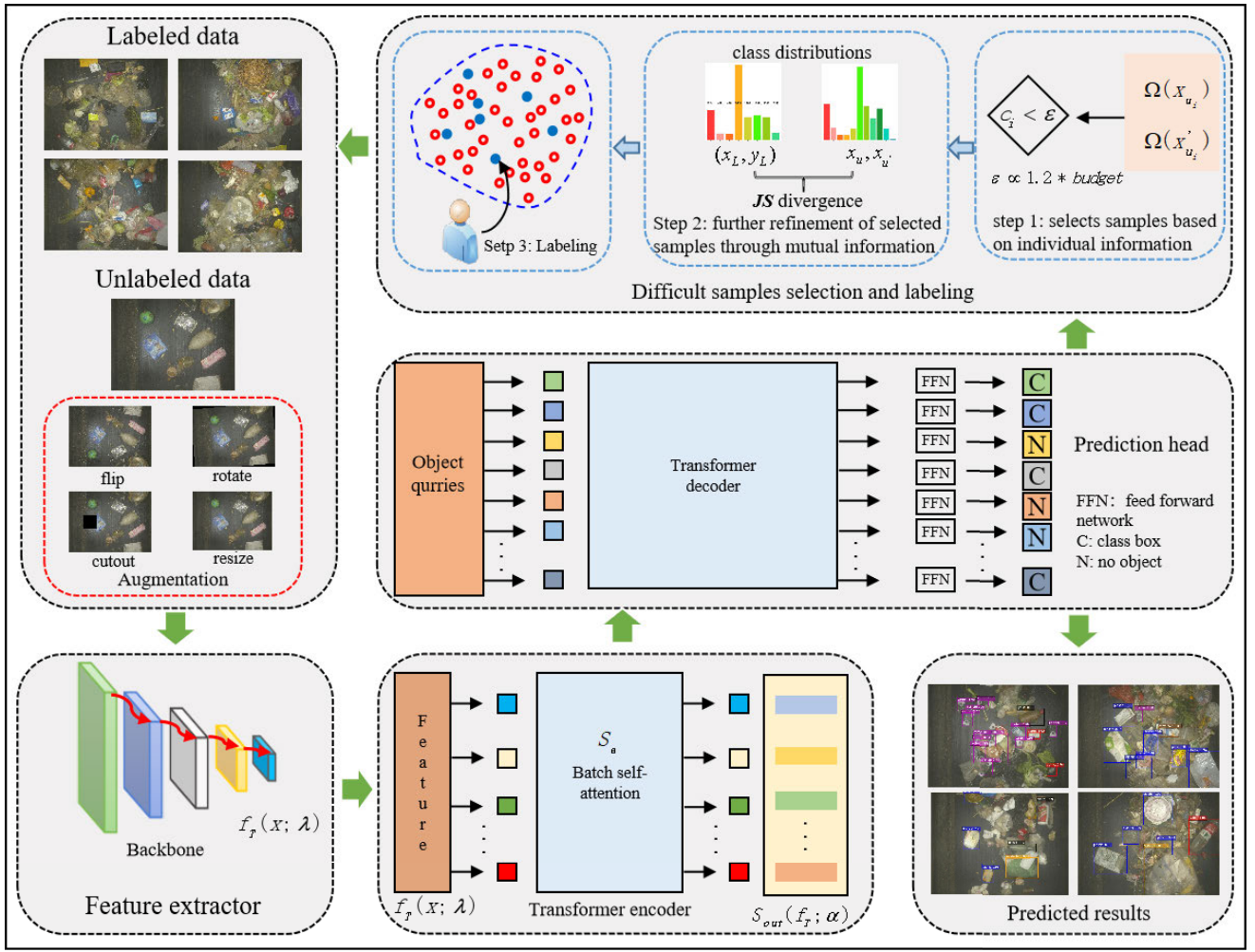


Fig. 6. AL-DETR algorithm framework overview.

here, S_a and softmax represent the output of the batch self-attentive layer and its activation function, and d_K denotes the dimension of the W_a^K and W_a^Q matrices, with the subscript a being shorthand for self-attention. We define $S_{out}(f_T; \alpha)$ as the output of the block, and α as encoder parameters. The following equation represents the process of passing S_{out} through the FFN

$$S_{out} = S_a + \odot(S_a W_{h1}) W_{h2}. \quad (2)$$

Here, W_h is the weight of the FFN, and $\odot$ represents the sigmoid activation function, such as ReLU in this article.

B. Selection of Challenging Samples

To choose samples with the highest uncertainty for labeling, we approach the problem using the following two criteria in comparison to existing methods.

- 1) We consider both bounding box and classification predictions in a single metric, combining box, and category uncertainties.
- 2) Our metric, in the unlabeled sample pool, is based on the average information of local regions rather than global images, focusing on the most informative regions. We will now elaborate on the detailed process of selecting challenging samples.

During the AL iteration, we introduce a detector with parameters ω and an image data augmentation method Aug (e.g., flipping, cropping, etc.). For an unlabeled image x_u , its augmented version is $x_u' = Aug(x_u)$. The bounding box and classification score predictions of the unlabeled image x_u can be represented as $\Omega(x_u, \omega)$, and through affine transformation invariance, these predictions can be mapped onto the augmented image x_u' . Hence, we denote the transformation of the i th prediction of x_u as the reference prediction, encompassing the prediction frame b_i and the prediction score s_i , by the following equation:

$$\{b_i\}, \{s_i\} = Aug(\Omega(x_u; \omega)). \quad (3)$$

Correspondingly, the bounding box and classification score predictions of the augmented image obtained directly from the detector can be expressed as

$$\{b'_j\}, \{s'_j\} = \Omega(x_u'; \omega). \quad (4)$$

The j in the equation denotes the prediction of the j th cycle round. We further consider the uncertainty between the reference prediction and the matching prediction. First, we match each reference prediction b'_i with a corresponding prediction score s'_j that includes the corresponding box b_i and the corresponding prediction scores of $\{b'_j\}, \{s'_j\}$. We select the corresponding box b'_i that has the maximum joint intersection

over union (IoU) with b_i . Here, s'_i is the prediction score of b'_i . The matching relationship between b_i and b'_i is as follows:

$$b'_i = \arg \max_{b'_j \in \{b'_j\}} \text{IoU}(b'_j; b_i). \quad (5)$$

Individual information is extracted from the image using a consistency measure between the reference prediction and the matching prediction. An initial selection pool is formed based on the prediction scores. For box regression, IoU is directly utilized, and the distance between the two types of probabilities is measured in terms of Jensen–Shannon (JS). To maintain the same trend as IoU, JS is inverted, i.e., 1-JS. The consistency of the i th prediction of an image c_i can be calculated as the sum of the consistency of the box B_i^b and the score B_i^s

$$\begin{cases} B_i^b = \text{IoU}(b_i, b'_i) \\ B_i^s = \frac{1}{2} [\max_{\theta \in s_i} (\theta_n) + \max_{\theta' \in s'_i} (\theta'_n)] (1 - \text{JS}(s_i \parallel s'_i)) \end{cases} \quad (6)$$

$$c_i = \frac{1}{2} (B_i^b + B_i^s) \quad (7)$$

where θ_n denotes the confidence level of the n th class. Clearly, $c_i \in [0, 1]$. In general, the smaller the consistency c_i , the greater the uncertainty of the prediction, which also implies that the sample is more informative. In accordance with the ascending order of the uncertainty c_i for all images within the unlabeled set, we select the top $1.2 \times \text{budget}$ of c_i as the candidate set for the second-step annotation selection. It is worth noting that the quantity of the “budget” varies in each round of active selection, and the threshold ε is positively correlated with it. Next, the mutual information between the samples in the initial unlabeled pool and the samples in the labeled pool is evaluated, determining the final selected unlabeled samples in the selected pool.

As illustrated in Fig. 1, there exists a significant disparity in the quantity distribution across each category. Therefore, we compute the JS divergence by evaluating the class distribution of the annotated examples and the previously selected unlabeled examples to choose samples that exhibit larger dissimilarities (i.e., those whose class distribution differs from the labeled pool). These selected samples constitute the final selection pool.

For the labeled pool, we aggregate all labels Y_L to represent the class distribution, as indicated by the following formula:

$$\begin{aligned} \Delta_L(Y_L) &= \text{Soft max}([\eta_1, \eta_2, \dots, \eta_m, \dots]^T) \\ \eta_m &= \sum_{y_L \in Y_L} \mathbb{E}(y_L = m) \end{aligned} \quad (8)$$

where m denotes the m th class within the dataset, and $\mathbb{E}$ represents the indicator function. Regarding the unlabeled samples X_u , we calculate the most confident class from both the original images and augmented images

$$\begin{aligned} \Delta_U(x_U) &= \text{Soft max}([\eta_1, \eta_2, \dots, \eta_m, \dots]^T) \\ \eta_m &= \max_{s_k \in \{s_k\}} \{\lambda_m \mid \lambda_m \in s_k\} + \max_{s'_k \in \{s'_k\}} \{\lambda'_m \mid \lambda'_m \in s'_k\} \end{aligned} \quad (9)$$

finally, we compute the JS divergence between these two sets and select samples with greater dissimilarity to form the

ultimate pool of challenging samples, which are subsequently handed over for expert annotation

$$F = \arg \max_{x_u \in X_L} [\text{JS}(\Delta_U(x_u) \parallel (\Delta_L(Y_L))]. \quad (10)$$

C. Loss Function for Object Detection Prediction

In the decoding step, the decoder generates a fixed-size set of N predictions, which is larger than the typical number of objects in the image. Scoring the category and detection frame information (e.g., location, size) of the predicted objects based on the GT poses a challenge. Our loss function establishes an optimal match between the predicted objects and the GT objects and subsequently optimizes based on object-specific losses such as bounding boxes. Let us denote the set of GT objects as $\hat{y} = \{\hat{y}_i\}_{i=1}^N$, where N denotes the set of predicted objects, and ϕ is filled if there is no target object. To find a mutual match between these two sets, we obtain a least-cost alignment according to the following formula:

$$\hat{\rho} = \arg \min_i \sum_i^N [-1_{\{t_i \neq \varphi\}} \hat{\rho}_{\rho(i)}(t_i) + 1_{\{t_i \neq \varphi\}} L_{\text{box}}(b_i, \hat{b}_{\rho(i)})] \quad (11)$$

where t_i is the target category label, and the probability of category t_i is defined as $\hat{\rho}_{\rho(i)}(t_i)$. b_i is a vector that includes the center coordinates of the GT boxes and its height and width relative to the image size, while the prediction boxes are defined as $\hat{b}_{\rho(i)}$. Each element i of the GT set is denoted as $y_i = (t_i, b_i)$. For the L_{box} , we use a linear combination of the $L1$ loss and the generalized IoU loss. The matching cost here takes into account both the category prediction and the similarity between the prediction frame and the GT boxes.

The final step is to define the loss function, which is similar to the loss of a typical object detector

$$L_d = \sum_{i=1}^N [-\log \hat{\rho}_{\rho(i)}(t_i) + 1_{\{t_i \neq \varphi\}} L_{\text{box}}(b_i, \hat{b}_{\rho(i)})]. \quad (12)$$

IV. EXPERIMENT AND ANALYSIS

We conducted experiments on the proposed dataset OD-KW, primarily divided into two parts to validate the performance of our proposed AL-DETR: the object detection performance of AL-DETR and the impact of data annotation quantity on detection performance. Subsequently, we provide a detailed discussion of the experimental evaluation metrics, experimental details, the effectiveness experiment of the created OD-KW dataset in object detection, the experiment on the impact of OD-KW dataset annotation quantity on object detection performance, and the real-world testing of the sorting platform constructed in this study.

A. Evaluation Criteria

In this article, we evaluate the performance of kitchen waste object detection using average precision (AP). Specifically, in a Cartesian coordinate system, with recall as the x -axis and precision as the y -axis, the area enclosed by the recall–precision

Algorithm 1 Pseudocode of the Proposed AL-DETR Framework

Input: Labeled images pool $X_L \in L$, unlabeled pool U , $x_u \in U \setminus L$, the batch of unlabeled images U_B .

Output: The final prediction bounding boxes and corresponding classification scores $y_i = (t_i, b_i)$

```

1: for  $U \neq \emptyset$  do
2:   Data augmentation of unlabeled sample  $x_u$  with batch
     size  $U_B$  results in  $x_{u'} = \text{Aug}(x_u)$ 
3:   Feed both  $X_L$  and  $U_B$  into the CNN for feature learning

4:   Encoding and computing batch self-attention  $S_a$  based
     on Eq. (1)
5:   Output  $S_{out}$  via FFN by Eq. (2) and then feed it into
     the decoder.
6:   Obtain the output from the decoder: labeled sample
     block  $S_{out-L}$ , unlabeled sample block  $S_{out-U}$  and unla-
     beled enhanced sample block  $S_{out-U'}$ .
7:   if  $U_B \neq \emptyset$  and  $\text{sum}(\text{select}) < \text{budget}$  then
8:     Calculate the uncertainty  $c_i$  of the  $S_{out-U}$  and  $S_{out-U'}$ 
        based on Eqs. (3)-(7)
9:     if  $c_i < \varepsilon$  then
10:      Compute the JS by Eqs. (8)-(10) and identify the
        samples to be labeled.
11:      Sample labels  $x_i \in U_B$ ,  $L \leftarrow x_i$ ,  $U_B \leftarrow U_B - x_i$ 
12:       $\text{sum}(\text{select}) \leftarrow \text{sum}(\text{select}) + 1$ 
13:    else
14:       $U \leftarrow U_B$ 
15:    end if
16:  end if
17:  Calculation of detection loss based on Eq. (11)-(12)
18: end for
19: Get the final prediction bounding boxes and corresponding
     classification scores.
  
```

(R-P) curve represents AP. mAP, which stands for the mean AP, is the average value of AP and is commonly used to assess the detection accuracy of algorithms. A higher mAP indicates better detection performance. The definitions of AP and mAP are as follows:

$$\text{AP} = \int_0^1 P(R) dR \quad (13)$$

$$\text{mAP} = \sum_{i=1}^N \frac{\text{AP}_i}{N} \quad (14)$$

where N represents the number of detected categories. Furthermore, for a comprehensive and detailed evaluation of detection accuracy, assessments are conducted for corresponding detection precision at different IoU thresholds ($\text{AP}_{[0.5:0.95]}$, AP_{75} , AP_{50}), as well as using the COCO evaluation mechanism to evaluate detection accuracy for objects of various scales (AP_S , AP_M , AP_L). Additionally, we have also employed other commonly used evaluation metrics for object detection, such

as average recall (AR), mean AR (mAR), and AR_S , AR_M , AR_L , which are standard protocols for COCO dataset.

B. Implementation Details

All experiments are implemented using the deep learning PyTorch 1.13.0 framework. The network employs adaptive moment estimation (AdamW) as the optimizer with a weight decay of 0.0002. AL-DETR is trained on an 11 GB NVIDIA GeForce 2080Ti GPU, CUDA 11.7, and CuDNN 8.5.0 GPU for 100 epochs with a batch size of 4. Specifically, the size of the input image is 640×480 . The learning rate adopts a stepwise decay. The initial learning rate is set to 0.000025, and the gamma is set to 0.8. The initial labeled ratio and the budget ratio of each AL cycle are set to 30% of all data and 25% of unlabeled data, respectively. The selection budget of the first stage at each cycle is $1.2 \times$ the overall budget to ensure that the selected samples reach the overall budget after the second stage of sample selection. In all AL cycles, we trained the DETR detector for 100 epochs with a batch size of 4. The numbers reported in the results are averages of three trials for each method and detector.

C. Object Detection Results on the OD-KW Dataset

To objectively affirm the efficacy of our proposed dataset and the merits of our approach, we conducted a comparison between our method and SOTA algorithms in object detection. We assessed the detection performance of the OD-KW dataset using both AL-DETR and established object detection benchmarks, namely YOLOv5s [33], YOLOv8s [34], Faster RCNN [35], and Deformable DETR [16]. The corresponding experimental results are detailed in Table I.

From Table I, it is evident that our proposed AL-DETR algorithm achieves the best performance in terms of AP_{75} , AP_L , and AR_L . Particularly, in AP_{75} , it outperforms the second-best method by 2.17%. For other metrics such as AP_M , AR_M , and mAP, the differences compared to the best methods are marginal, not exceeding 1%. This indicates that AL-DETR can focus on the information differences in local features of the images. By actively selecting challenging examples through the AL method and integrating the transformer approach, it enhances the feature learning ability, resulting in relatively balanced detection performance across various garbage categories.

Following the COCO evaluation criteria, this study conducted a quantitative analysis of the OD-KW dataset, categorizing objects into small, medium, and large classes, as depicted in Fig. 7. Notably, large objects account for 87.5% of the dataset, making the APL metric more meaningful for comparison. Combining the experimental results in Table I, our method outperforms other approaches significantly, particularly in the APL metric. Specifically, we surpass the current leading YOLOv8s algorithm from the YOLO series by 13.5%. In comparison to Deformable DETR, which is based on the same DETR methodology as this article, our method exhibits a superiority of over 9.23%. Furthermore, the ARL metric also demonstrates an improvement of at least 1.3% compared to other methods. This suggests that our method,

TABLE I

COMPARATIVE WITH YOLOv5s, YOLOv8s, FASTER RCNN, AND DEFORMABLE DETR UTILIZING RESNET-50 ON THE VALIDATION SET. RESULTS ARE AVERAGED ACROSS ALL THREE FOLDS WITH 100 EPOCHS. BOLD VALUES INDICATE TOP-PERFORMING RESULTS. mAP AND mAR REPRESENT MAP AND MAR, RESPECTIVELY, CALCULATED FROM IOU = 0.5 TO IOU = 0.95

Model	AP_{50}	AP_{75}	mAP	AP_S	AP_M	AP_L	mAR	AR_S	AR_M	AR_L	FPS
YOLOv5s	66.60	60.33	55.43	17.33	54.77	37.83	77.60	32.83	77.20	75.67	50.00
YOLOv8s	69.50	63.30	59.70	23.40	59.70	41.80	77.20	35.40	75.80	71.20	111.99
Faster RCNN	74.70	64.70	57.50	14.80	58.80	42.97	71.53	20.97	70.13	69.93	12.15
Deformable DETR	74.02	61.90	53.00	16.50	54.33	47.20	72.23	17.33	71.57	70.73	9.64
AL-DETR	73.00	66.87	58.90	19.00	59.33	56.43	76.63	28.33	75.23	76.90	27.03

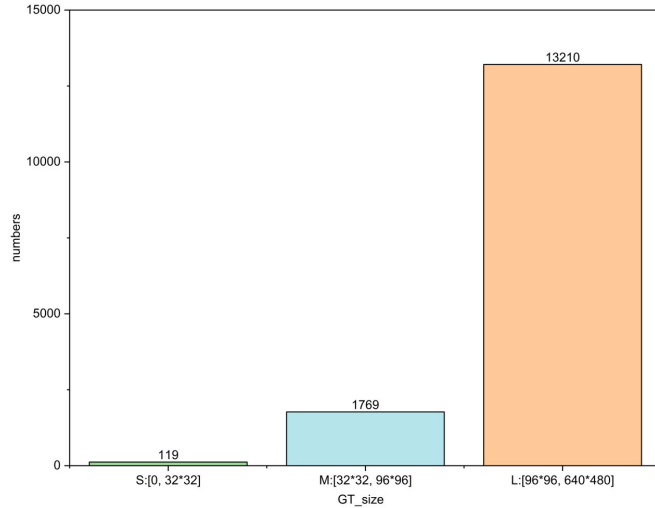


Fig. 7. Distribution of objects based on size categories (S , M , L), following the COCO evaluation criteria definition in the OD-KW dataset.

focusing on batch feature learning and calculating sample correlations, excels in selecting the most informative examples for annotation. It demonstrates robust non-local feature learning capabilities for larger kitchen waste objects, highlighting the advantages of AL in object detection. It is noteworthy that our method excels in the detection of large objects compared to other approaches. However, in terms of the FPS metric, although our method outperforms Faster RCNN and Deformable DETR, it lags behind YOLO-based methods. Nevertheless, on the OD-KW dataset, primarily focusing on large object detection, our method undeniably exhibits significant superiority. Furthermore, deploying the algorithm onto the designed experimental sorting platform, considering the control response and execution speed of the robotic system, the proposed algorithm's FPS is fully capable of meeting the detection requirements. Therefore, striking a balance between detection accuracy and speed, our method achieves a satisfactory outcome. Visualization in Fig. 8 presents the real annotations of kitchen waste images and the results of object detection using five algorithms, emphasizing the precision of detection boxes and category judgments achieved by the proposed method.

D. Ablation Studies

To assess the impact of the two-stage AL module and the object detection DETR module on the proposed AL-DETR

TABLE II
ABLATION STUDIES ON DIFFERENT STAGE STRATEGIES ABOUT mAP(%)

Model	30%	47.50%	65%	82.50%	100%
Random-DETR	46.87	51.43	55.60	57.70	59.77
AL1-DETR	47.23	52.20	55.33	57.03	59.53
AL2-DETR	46.93	50.67	54.63	57.43	59.00
AL-DETR	47.33	53.40	55.97	58.28	59.63

TABLE III
ABLATION STUDIES ON DIFFERENT STAGE STRATEGIES ABOUT AP_{50} (%)

Model	30%	47.50%	65%	82.50%	100%
Random-DETR	60.20	65.10	69.37	71.60	73.47
AL1-DETR	60.73	65.27	69.13	71.10	73.07
AL2-DETR	60.60	63.57	68.23	71.03	73.03
AL-DETR	60.95	66.82	69.47	72.25	73.50

algorithm, we conducted ablation experiments considering scenarios without an AL module, with only AL in the first stage (AL1-DETR), with only AL in the second stage (AL2-DETR), and with both stages of AL querying challenging samples (AL-DETR). The experiments include random-DETR as a baseline. The results for mAP (IoU = 0.5 to IoU = 0.95) and AP_{50} (IoU = 0.5) are presented in Tables II and III, respectively.

Based on the experimental outcomes, it is evident that DETR, in the absence of AL, exhibits comparatively lower mAP performance, particularly under conditions of limited labeled samples, when contrasted with DETR integrated with the AL method. AL1-DETR, focusing on querying challenging samples solely in the first step, and AL2-DETR, concentrating on challenging samples exclusively in the second step, rely exclusively on the consistency between predictions from original and augmented images, as well as the class distribution distance (measured by JS divergence) between the selected pool and the annotated pool. Due to the absence of mutual information between the two steps, when compared with AL-DETR, which incorporates both steps simultaneously, the mAP is lower for the former under the condition of an equal number of labeled training samples. Consequently, the experimental results affirm the efficacy of the module integration proposed in this study. The changes in mAP and AP_{50} for different fusion methods are visually depicted in the curve chart in Fig. 9.



Fig. 8. Object detection performance comparison. Visualization results from left to right: original image, GT, YOLOv5s, YOLOv8s, Faster RCNN, Deformable DETR and AL-DETR, respectively. (a) Image. (b) GT. (c) YOLOv5s. (d) YOLOv8s. (e) Faster RCNN. (f) Deformable DETR. (g) AL-DETR.

In addition, we analyzed the impact of different fusion methods on the experiment from the perspective of specific target categories. The two-stage sampling strategy in this study significantly improves the detection performance of challenging and low-sample categories. Categories with initial sample mAP below 40%, such as “carton” and “glass” shown in Fig. 10, are considered challenging sample categories. Additionally, based on the dataset’s different category quantities and ratios shown in Fig. 2, categories with proportions below 10%, such as “metal,” “shoe,” and “glass,” are considered low-sample categories. Fig. 10 illustrates the AP value improvements for each category after the first two rounds of active selection under three different stage selection strategies. Fig. 10(a) shows results using only the first-stage selection strategy, with similar improvements for each category, close to the overall mAP improvement of 4.87% and 3.13%. Fig. 10(b), using only the second-stage selection strategy, tends to improve the detection performance of challenging sample categories and low-sample categories

such as “shoe” ($\uparrow 10.04\%$), “metal” ($\uparrow 7.27\%$), and “glass” ($\uparrow 8.67\%$). This indicates that the first stage equally focuses on individual information for each target, while the second stage focuses on the mutual information of the samples. Fig. 10(c), using both stages to select samples, shows a more balanced and stable overall AP improvement for each category compared to Fig. 10(b) using only the second stage. Compared to Fig. 10(a), using only the first stage, the improvement in challenging sample categories and low-sample categories is more significant. It simultaneously considers both individual information and mutual information in sample selection.

E. Impact of Data Annotation Volume on Detection Performance

AL mitigates the need for extensive annotated data by focusing on challenging examples that pose difficulties for the model. This study incorporates an uncertainty-based AL strategy into the DETR method and conducts experiments

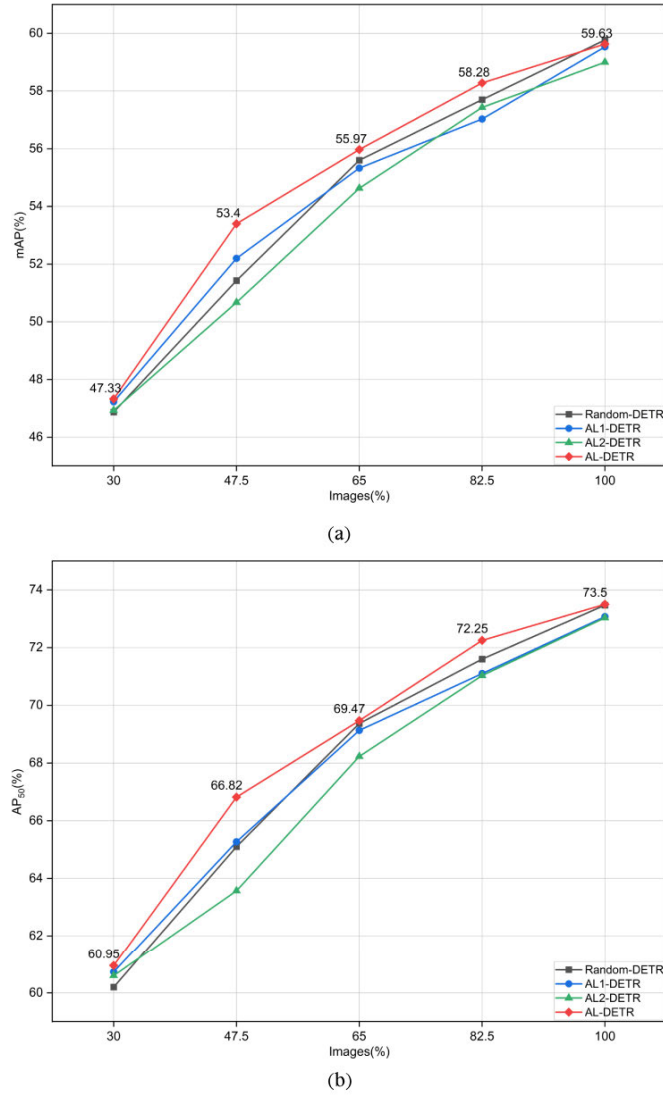


Fig. 9. (a) Comparison of performance (mAP) in active object detection methods with different stage strategies. (b) Comparison of performance (AP₅₀) in active object detection methods with different stage strategies.

across varying annotation volumes of the OD-KW dataset. Additionally, we compare the integration of DETR with Faster RCNN against random selection, two specific SOTA AL methods in object detection (SSM [25] and LS + C [24]), and a task-agnostic AL method (VAAL [26]). The results of these experiments are detailed in Table IV.

The experimental results indicate that the proposed AL-DETR outperforms other AL methods when the training data annotation volume is relatively low (30%, 47.5%, 65%, 82.5% annotation volume). The obtained mAP values consistently surpass those of other methods. Moreover, the AL approach integrated with DETR performs better than Faster RCNN, highlighting the transformer's ability to focus on more image details in object detection, thereby achieving more accurate target positions and categories. On the entire annotated training set, the difference with the random-DETR method is minimal, which is related to the total volume of our annotated data. As the training data increases, the model's

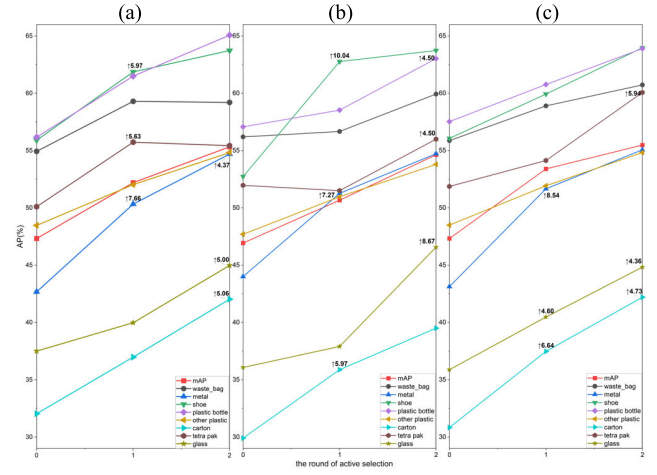


Fig. 10. Changes in mAP rates for each category after two rounds of AL sample selection training under AL1-DETR, AL2-DETR, and AL-DETR (denoted as First, Second, First and Second, respectively) strategies. The numbers in the figure represent the growth amounts of the top 3 categories in mAP after sample selection in each round.

TABLE IV

DETECTION mAP (%) RESULTS FOR DIFFERENT ANNOTATION VOLUMES ON OD-KW USING VARIOUS AL STRATEGIES WITH DETR (RESNET-50) AND FASTER RCNN (RESNET-50). BOLD VALUES INDICATE THE TOP-PERFORMING RESULTS

Model	30%	47.50%	65%	82.50%	100%
Random-DETR	46.87	51.43	55.60	57.70	59.77
LS+C-Faster RCNN	40.30	47.47	50.93	53.57	55.37
LS+C-DETR	44.53	49.03	51.97	55.27	57.20
SSM-Faster RCNN	40.67	46.97	50.60	53.13	55.17
SSM-DETR	44.37	49.90	52.80	55.30	57.27
VAAL-Faster RCNN	40.60	46.63	50.43	53.33	55.37
VAAL-DETR	43.73	49.47	53.00	54.57	56.60
AL-Faster RCNN	40.53	48.17	52.10	54.23	55.43
AL-DETR	47.33	53.40	55.97	58.28	59.63

TABLE V

AP₅₀ (%) USING DETR (RESNET-50) ON OUR DATASET. BOLD VALUES INDICATE THE TOP-PERFORMING RESULTS

Model	30%	47.50%	65%	82.50%	100%
Random-DETR	60.20	65.10	69.37	71.60	73.47
LS+C-DETR	57.83	63.00	66.27	69.63	71.67
SSM-DETR	58.27	64.30	67.67	69.83	72.27
VAAL-DETR	56.87	62.93	66.83	68.40	70.70
AL-DETR	60.95	66.82	69.47	72.25	73.50

accuracy improves. The curves in Fig. 11 intuitively display the mAP values obtained by different methods at various levels of training annotation volume.

In addition to mAP (IoU = 0.5 to IoU = 0.95), we conducted comparative experiments on the detection performance with different AL methods integrating DETR, focusing specifically on the AP₅₀ (IoU = 0.5) metric, as shown in Table V.

The curves in Fig. 12 illustrate the consistent top performance of our proposed method, represented by the red curve.

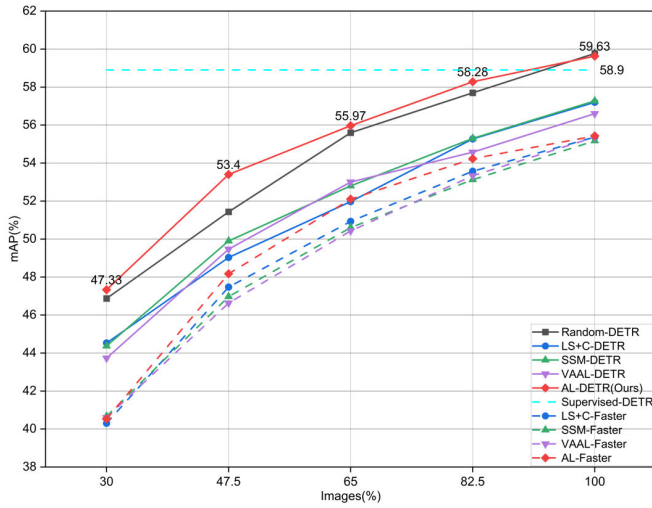


Fig. 11. Comparison of mAP performance using different AL approaches integrated with DETR and Faster RCNN.

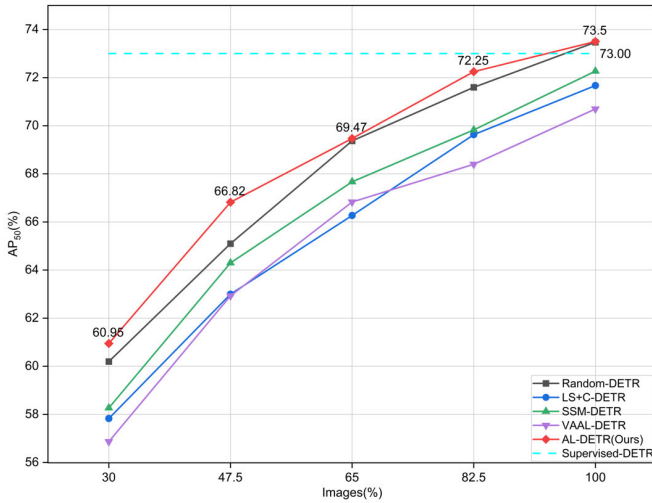


Fig. 12. Comparison of AP₅₀ performance using different AL approaches integrated with DETR and Faster RCNN.

This indicates the overall superiority of our approach in terms of the AP₅₀ evaluation metric. The scenario of a conveyor belt with visually similar and cluttered kitchen waste can pose challenges for the model, leading to misjudgments. The AL strategy implemented in this article focuses on selecting examples with high uncertainty for annotation, facilitating the model's learning of complex background images.

Moreover, as the kitchen waste target pixels under consideration in this study are predominantly large, a focused investigation into the detection of large objects becomes essential. Adhering to the evaluation metrics of the COCO dataset, we conducted detection experiments specifically on large objects (pixels larger than 96×96) within the OD-KW dataset. The comparative methods continue to include different AL approaches integrating DETR and Faster RCNN. The experimental results are detailed in Table VI.

From the experimental results, it is evident that the proposed method achieved the best performance at the minimal training sample annotation volume of 30%, surpassing the second-best

TABLE VI
AP_L(%) USING DETR (RESNET-50) ON OUR DATASET. BOLD VALUES INDICATE THE TOP-PERFORMING RESULTS

Model	30%	47.50%	65%	82.50%	100%
Random-DETR	37.07	37.73	42.83	44.83	48.87
LS+C-Faster RCNN	25.93	30.40	34.37	36.60	38.70
LS+C-DETR	35.50	41.57	40.50	45.53	50.77
SSM-Faster RCNN	26.83	29.80	33.63	37.63	38.83
SSM-DETR	39.67	40.67	50.03	50.77	45.50
VAAL-Faster RCNN	25.97	29.00	32.87	37.47	39.23
VAAL-DETR	34.57	53.43	49.80	52.90	51.87
AL-Faster RCNN	25.70	33.70	36.80	38.40	39.70
AL-DETR	41.63	43.90	48.10	54.93	51.67

method by 2%. Additionally, at an annotation volume of 82.5%, it attained the highest value of 54.93% among all methods, indicating that the AL-DETR method can indeed detect larger kitchen waste objects under conditions of limited annotated samples. However, it is noteworthy that training with all annotated samples led to a decrease in the AP_L value. One possible reason is the limited annotation volume in our dataset, resulting in overfitting during training. This is an aspect we aim to address in future improvements, such as annotating more data and optimizing the network model.

F. Experimental Evaluation of Sorting System in Practical Testing

In the implemented kitchen waste sorting platform, depicted physically in Fig. 13(a), we conducted practical tests to validate the effectiveness of our algorithms in the control software and robotic sorting. The integrated algorithms in the XRobot control software successfully predicted the location of the garbage and generated corresponding detection boxes in real time, as demonstrated in Fig. 13(b). The sorting process involved transmitting category information to the robotic arm, which employed different methods based on the type of garbage. For items like plastic bottles, aluminum cans, shoes, and tetra paks, the robotic arm utilized a gripping method, as illustrated in Fig. 13(c). On the other hand, garbage categories like plastic bags and rigid paper were sorted using an adhesive method, as shown in Fig. 13(d). It is worth noting that the gripper of our robotic arm, equipped with force tactile sensors, is designed to handle fragile items like glass bottles. The control software allows the adjustment of gripping force to prevent the shattering of glass bottles during the sorting process. In terms of the gripping and suction force control settings for the gripper, we have set a threshold of 20N for adjusting the grasping force when recognizing glass bottle targets. To streamline the process, we have established fixed grasping forces of 13N for other garbage categories, respectively. For items like waste bags and cartons, pneumatic suction is utilized with pressure ranging from -60 to -80 kPa for sorting. In order to evaluate the effectiveness of the robotic sorting system, we have conducted 300 sorting experiments, and an average success rate of 82.7% for gripping and suction was achieved. The demonstration video

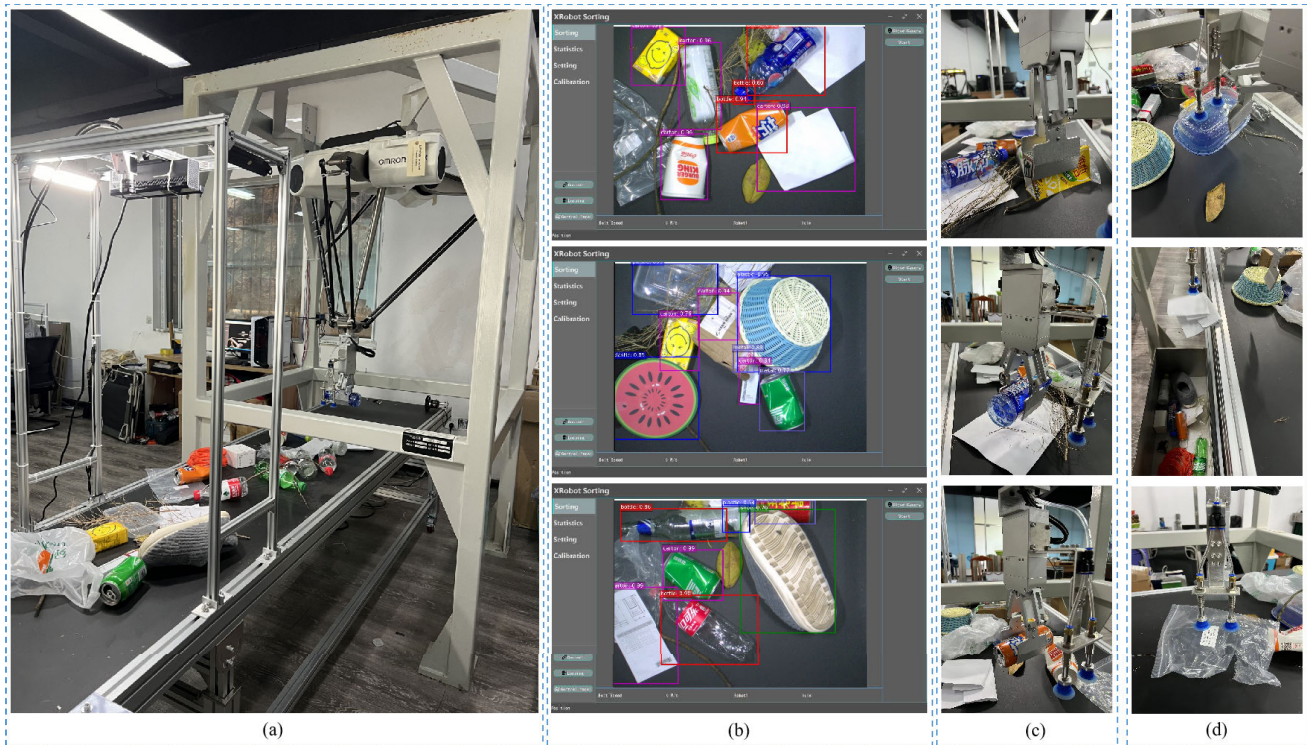


Fig. 13. (a) Physical platform for kitchen waste sorting. (b) Real-time kitchen waste detection and generation of position boxes with predicted categories by XRobot. (c) Robotic sorting of kitchen waste using gripping method. (d) Robotic sorting of kitchen waste using adhesive method.

for the on-site experiment can be found at the following link: <https://github.com/seedahi/Robotic-Intelligent-Waste-Sorting>.

V. CONCLUSION

This article systematically addresses the challenges of diverse categories, significant morphological variations, and the unfavorable conditions of traditional manual sorting methods in the context of kitchen waste. First, we construct the OD-KW dataset, encompassing eight categories for kitchen waste object detection. Subsequently, leveraging AL to reduce data annotation costs and capitalizing on the advantages of DETR in object detection, we propose AL-DETR. Experimental results demonstrate that our algorithm achieves SOTA performance in kitchen waste object detection with a limited amount of annotated data. Additionally, we establish a kitchen waste sorting platform and develop the XRobot software for detection and robot control in sorting. In practical tests, our algorithm detects targets, and various types of waste are handed over to the robotic arm for grabbing or suction, achieving automated sorting and efficient recycling of kitchen waste.

In our future work, we will focus on further improving the accuracy of kitchen waste object detection. This includes refining the robotic manipulator arm structure, selecting more precise force/tactile sensors, and designing grippers with a larger aperture angle to enhance their capability to grasp various types of objects. The ultimate goal is to achieve faster and more reliable automated sorting in industrial scenarios.

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